# AirtelSync • Solution Architecture

Project Udyam Digital · Accel Business · digital-first MSME discovery, bundling, KYC and activation · YTL Hackathon 2026, Track A

Numbered data flows ① – ⑤ (key below) — Request / response — - - Read / write to data layer

Mocked in prototype, real-API shaped

## USERS & CHANNELS

**MSME owner**

Shop, clinic, office, small chain or factory owner on a mobile phone.

Types or speaks in English, Hindi, Hinglish or another Indian language; the app follows their language every turn.

Installs nothing: opens a web link (installable PWA).

Gets status updates in-app and on WhatsApp.

**Accel staff & partners**

Sales and KYC reviewers; partner and field agents in assisted mode.

## WEB APP React PWA

① **Advisor chat**
Text or voice, EN / हिंदी. ~5 questions; an optional GSTIN prefills details.

② **Bundle cards & quote builder**
Essential / Recommended / Complete tiers with reasons. Toggle items, pick the term, see price incl. GST, accept.

③ **Smart KYC**
Consent, camera upload or DigiLocker, instant fix-it feedback, e-sign.

④ **Order tracker**
Live timeline to activation; delays and reroutes in plain words.

⑤ **Ops console**
Lead scores & routing, re-engage list, KYC review queue, assisted mode.

`POST /advisor/sessions/{id}/messages`

`tiers + reasons`

`POST /checkout/{sid}/quote · accept`

`POST /kyc/cases/{id}/consent · documents · verify · esign`

`GET /orders/{id}/track`
`POST mandate`

`GET /console/leads · reengage · kyc-reviews`

## APPLICATION SERVICES FastAPI (Python) · one process · routes under /api

**① Advisor**
advisor/ — orchestrates the conversation

Language detect → LLM with tool calling → Output guard → Reply · Scripted fallback

The LLM talks; tools do the work: `lookup_gstin · update_profile · recommend_bundle · check_product · handoff_to_human`.

Guard: every ₹ amount and product ID must come from a tool result (one retry). A failed or slow turn (35 s) is answered by the rule-based EN / HI / Hinglish advisor.

`tools: recommend_bundle · check_product`

**② Product engine**
engine/ — deterministic, no AI

Eligibility rules → Pricing → 3-tier recommender

Catalogue rules: city tier, staff, sites, active GSTIN, required companions. Price falls with term; 5 % off for 3+ categories; micro firms' setup waived on 12+ months; GST 18 %. Tiers carry reason codes.

`chosen tier`

**Checkout**
checkout/

Edit items & term → Server re-prices & validates → Order: QuoteAccepted

The browser never sets a price: each change is re-priced and re-checked by the engine.

`order needs KYC`

**③ Smart KYC**
kyc/ — automated, explainable

DPDP consent → Upload / DigiLocker → OCR (Tesseract) → Parse & repair IDs → 6 check layers → Verdict → e-Sign

Checks: quality (blur, skew) · format (GSTIN / PAN / Udyam, checksum repair) · completeness · registry · cross-document consistency · recency. Verdict: **approved** · **needs action** (owner fixes) · **review** (staff queue) · **blocked**, each with reasons.

`KYCVerified`

**④ Orders & notifications**
orders/ — fulfilment state machine

QuoteAccepted → KYCVerified → MandateSet → InstallScheduled → Installed → Activated

No fibre at the site → AirFiber reroute with new price. Missed install SLA → delay alert. Each step queues an EN / HI message in the notification outbox.

`status · SLA · outcomes`

**⑤ Ops console**
console/ — role-based (sales · partner)

Propensity model → Route: digital · partner · field · Re-engage list · KYC reviews · Assisted mode

Logistic regression on lead features (trained Apr–May, tested June: AUC 0.70, top-decile lift 2.45×). Assisted mode needs a code the owner shares.

`prompt + tool definitions ⇄ function calls / reply`

`GSTIN → business profile prefill`

`verify GSTIN / Udyam · pull documents · e-sign OTP`

`UPI mandate`

`notify owner`

## EXTERNAL SERVICES & AI

**Google Gemini API** LIVE

`gemini-2.5-flash` → falls back to `flash-lite` → `flash-latest` when busy or rate-limited (429 / 503), with retries.

Claude supported as an alternative. Only chat text and tool results are sent, never KYC documents.

**Government & identity** MOCK

- **GSTIN registry** — names, status (Active / Cancelled / Suspended), address
- **Udyam registry** — enterprise class (micro / small / medium), activity
- **DigiLocker** — pull issued documents with the owner's consent
- **Aadhaar e-Sign** — OTP signature on the service agreement

Served by `/api/mock/*` from the case registry files, in the real APIs' shape so each can be swapped for production.

**Payments & messaging**

- **UPI Autopay mandate** mock — monthly debit set-up
- **WhatsApp** via Twilio sandbox (optional) — EN / HI status updates

**Production hand-offs**

Install scheduling, billing and CRM would subscribe to the same order events; not built in the prototype.

`read / write`

## Data layer

One store module; every write timestamped. Raw documents never enter the operational store.

**Operational store** `store.py · SQLite`

`advisor_sessions · orders · kyc_cases · consents · notifications · audit`

**KYC vault** `kyc/vault.py`

Uploaded images encrypted at rest (Fernet), deleted once no longer needed (≤ 7 days only while a staff review is pending). Extracted fields and a hash remain; screens show masked IDs.

**Reference data** `data/*.csv`

`product_catalogue · smb_leads · mock_gstin_registry · mock_udyam_registry`

Catalogue drives the rules and prices; leads train the propensity model.

## PRIVACY & SECURITY BY DESIGN

- DPDP consent ledger: purpose-bound, revocable
- Chat never asks for Aadhaar, PAN or bank details; IDs typed in chat are masked
- Encrypted vault and deletion of raw images after the decision
- Audit log of consent, document, verification, review and assisted-mode events
- Role-based console: partners see only leads routed to them
- The AI cannot invent prices or products (output guard)

### ① Discover

1. Owner chats or speaks in their language.
2. Advisor sends the turn to Gemini, which calls tools: GSTIN lookup prefills the profile; the engine checks eligibility.
3. Guard checks the draft reply; tier cards appear after ~5 questions.

### ② Quote & accept

1. Owner picks a tier, toggles items, chooses the term.
2. Checkout re-prices each change through the engine (rules, discounts, GST).
3. Accept creates the order in state **QuoteAccepted**.

### ③ Verify (Smart KYC)

1. Owner gives consent, then uploads documents or uses DigiLocker.
2. OCR and ID checks against the registries produce a verdict with reasons; unclear cases go to staff review.
3. E-sign moves the order to **KYCVerified**.

### ④ Activate & track

1. UPI mandate, then installation is scheduled and completed.
2. Reroutes and delays are handled in the state machine.
3. Each step notifies the owner in-app and on WhatsApp, ending at **Activated**.

### ⑤ Learn & route

1. The console scores every lead and routes it: digital, partner or field.
2. Stalled leads go on a re-engage list; staff clear the KYC review queue.
3. Partners can help an owner in assisted mode, with the owner's code.

Prototype: React + TypeScript PWA · FastAPI · SQLite · Gemini · Tesseract OCR — github.com/AgileSeagull/AirtelSync

Accel Business is the case's fictional operator. Registries, DigiLocker, e-Sign and UPI are mocks for the prototype.